

Aditi Nayak

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LinkedIn | GitHub | Portfolio

Summary

Software engineer working on backend systems and AI applications. I build REST APIs, background job processing, and vector search tools using Python, FastAPI, Django, and PostgreSQL. Good foundation in authentication, database concurrency, and automated deployment pipelines. Combines a strong Data Structures and Algorithms foundation with hands-on experience building, testing, and containerizing distributed applications.

Education

Lakhmi Chand Institute of Technology

2021 – 2025

B.Tech in Computer Science and Engineering | CPI: 8.42

Technical Skills

- **Languages:** Python, SQL, JavaScript
- **Backend Frameworks:** FastAPI, Django REST Framework, Flask, REST APIs, SQLAlchemy, Alembic, Pydantic, Celery
- **AI and Retrieval:** Multi-Agent Systems, RAG Architecture, pgvector, LangChain, Hugging Face, Gemini API, OpenAI
- **Databases:** PostgreSQL, MySQL
- **DevOps and Testing:** Docker, GitHub Actions, Render, Pytest, Gunicorn, Uvicorn
- **Security:** JWT, OAuth2, RBAC, Rate Limiting (SlowAPI), Bcrypt, HMAC

Projects

Webhook Delivery Platform

[Link](#)

Tech Stack: Python, Django REST Framework, Celery, Redis, PostgreSQL, Docker

- Designed an event delivery backend with Django and PostgreSQL, using HMAC SHA256 signatures to verify payloads.
- Sent background tasks to a Celery and Redis worker queue with retry logic to handle target endpoint downtime.
- Containerized the project with Docker and Gunicorn to keep all API endpoints fast and non-blocking during background runs.

Task Management REST API

[Link](#)

Tech Stack: FastAPI, Python 3.13, PostgreSQL, SQLAlchemy, Alembic, Docker, GitHub Actions, Streamlit

- Built a production REST API in FastAPI and PostgreSQL, setting up Role-Based Access Control across Admin, Manager, and Viewer roles.
- Added JWT access tokens and server-side refresh tokens, using bcrypt password hashing and rate limiting on login routes.
- Handled concurrent data updates using version fields and automated testing and deployment with GitHub Actions and Docker on Render.

DocuMind AI Document Chat Application

[Link](#)

Tech Stack: FastAPI, React, pgvector, Google Gemini API, GitHub Actions, Pytest

- Developed a multi-tenant RAG application using FastAPI, React, and Gemini APIs to read PDFs and run vector search with pgvector.
- Kept user data separate and protected using JWT authentication and account-scoped queries across all endpoints.
- Set up CI pipelines using Pytest and GitHub Actions with pgvector service containers, running over 60 automated tests.

Achievements

- **Data Structures and Algorithms:** Solved 200+ coding problems on LeetCode covering dynamic programming, graphs, and array techniques.
- **CI/CD and Deployment:** Set up automated test and deployment pipelines using GitHub Actions and Render to keep live APIs running.
- **Engineering Standards:** Completed a virtual software engineering internship learning git branch workflows, code reviews, and clean API design.